

Basics of Probability and Statistics

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Events and Sample Spaces



Coin Toss - $\{H, T\}$



Die Roll - $\{1, 2, 3, 4, 5, 6\}$



Weather of Day - $\{\text{Sunny, Windy, Cloudy, Rainy}\}$



Discovery of minerals at a place - $\{\text{Yes, No}\}$

Intersection

$$P(\cap_{i=1}^n A_i) = P(A_1 A_2 \dots A_n)$$

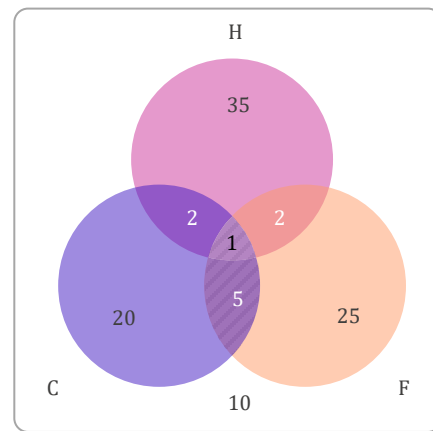
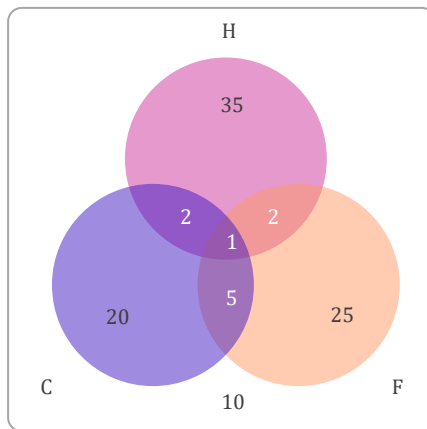
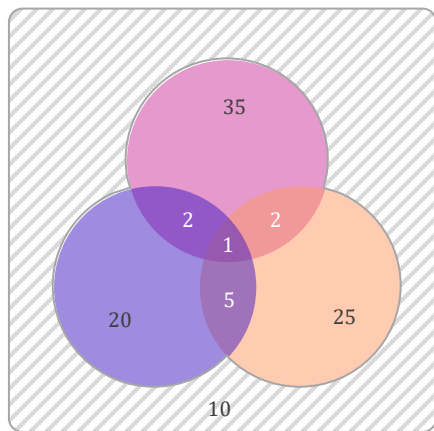
Union

$$P(\cup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i} \sum_{< j \leq n} P(A_i A_j) + \sum_{1 \leq i} \sum_{< j} \sum_{< k \leq n} P(A_i A_j A_k) - \dots - (-1)^n P(A_1 A_2 \dots A_n)$$

Complement

$$P(A_i^c) = 1 - P(A_i)$$

Algebra of Events



$$\begin{aligned}
 P(\text{none}) &= P((C \cup H \cup F)') = 1 - P(C \cup H \cup F) \\
 &= 1 - [P(C) + P(H) + P(F) - P(CH) - P(CF) - P(HF) + P(CHF)] \\
 &= 1 - [0.28 + 0.4 + 0.33 - 0.03 - 0.06 - 0.03 + 0.01] \\
 &= 0.1
 \end{aligned}$$

$$P(CF) = 0.06$$

Mutual Exclusivity and Exhaustivity

Mutually Exclusive Events

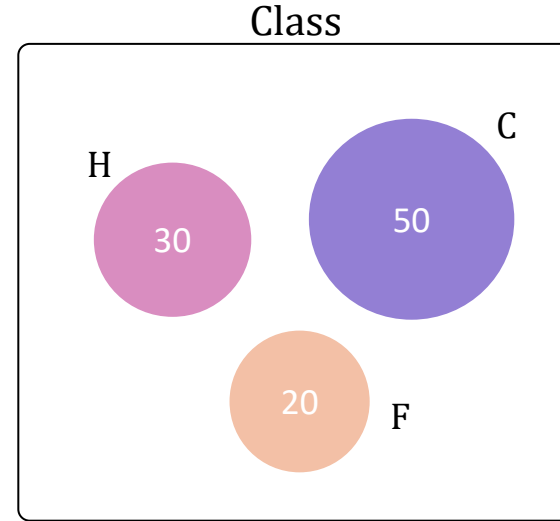
$$P(E_i E_j) = 0 \quad \forall i \text{ and } j$$

$$P(\cup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i)$$

Exhaustive Events

$$P(\cup_{i=1}^n E_i) = 1$$

Example:-



Conditional Probability

Probability of Event A happening given that Event B has already happened.

$$P(A/B) = \frac{P(AB)}{P(B)}$$

Conditional Probability

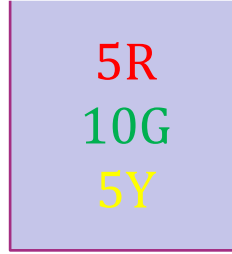


1

$$P(R|1) = \frac{P(R,1)}{P(1)}$$

$$= \frac{10/60}{20/60}$$

$$= 10/20 = 0.5$$



2

$$P(R|2) = \frac{P(R,2)}{P(2)}$$

$$= \frac{5/60}{20/60}$$

$$= 5/20 = 0.25$$



3

$$P(R|3) = 5/20 = 0.25$$

Independence

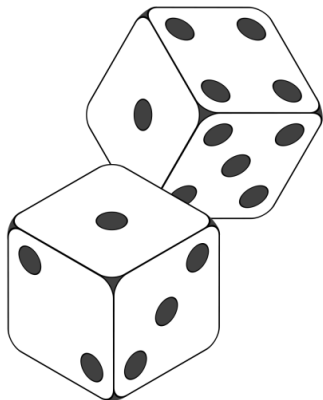
$$P(E|F) = P(E)$$

$$\text{since, } P(E/F) = \frac{P(EF)}{P(F)}$$

$$\text{thus, } P(EF) = P(E) P(F)$$

Independence

Outcome of rolling 2 dice.



Dependence

Weather given season.



Random Variables(RV)

Discrete Case

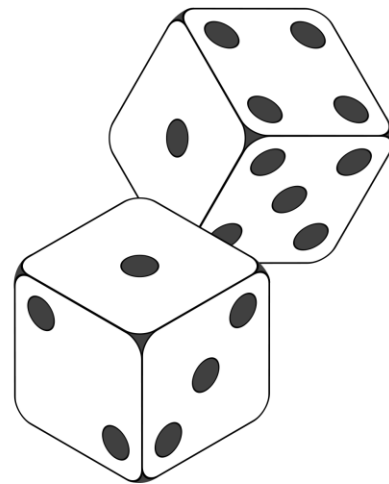
$X = \text{sum of 2 die roll}$

$$\text{Set}(X=2) = \{(1,1)\}$$

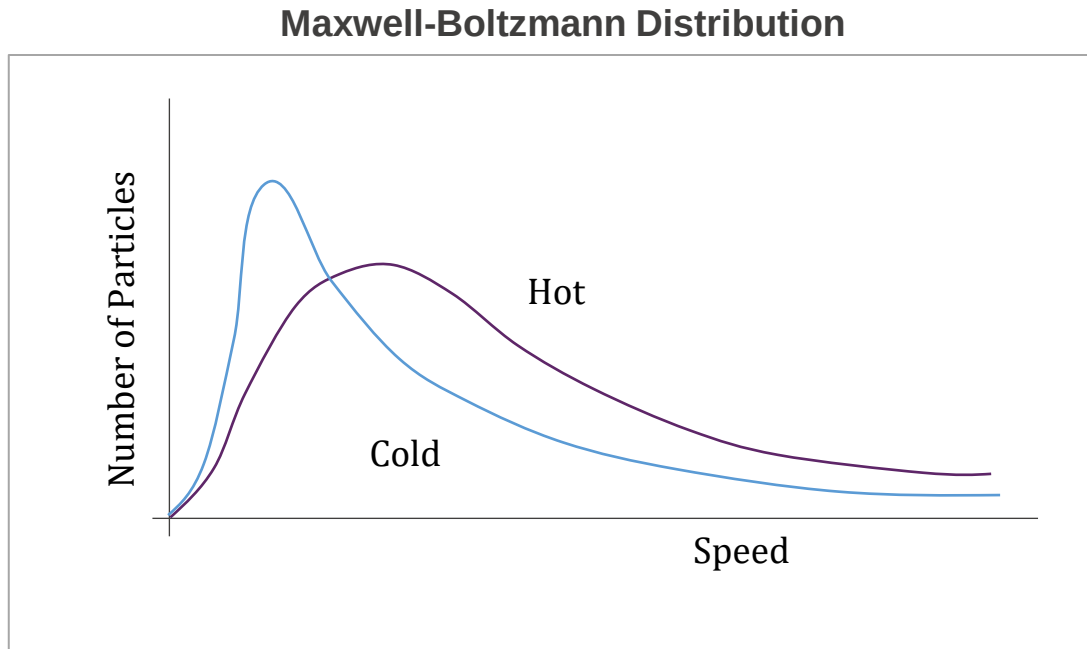
$$\text{Set}(X=4) = \{(1,3), (2,2), (3,1)\}$$

$$\text{Set}(X=7) = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$\text{Set}(X=11) = \{(5,6), (6,5)\}$$



Continuous Case

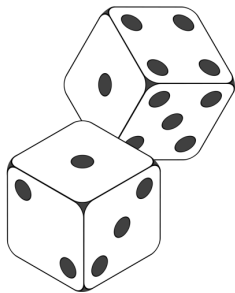


Associated Functions with RV

Probability Mass Function(Discrete only)

$$P(X=x) = p_X(x)$$

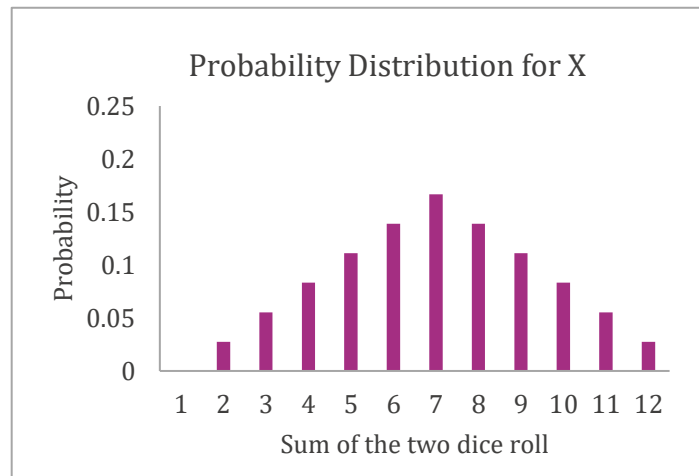
Example:-



Sum of 2 Die roll

$$P(X=2) = 1/36$$

$$P(X=7) = 6/36$$



Probability Density Function (Continuous)

$$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a) = F_X(b) - F_X(a)$$

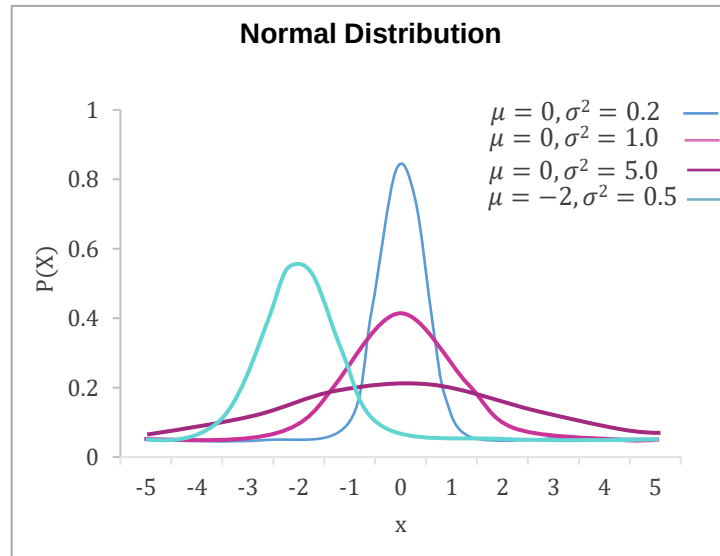
Density of X between a and b

$$P(a \leq X \leq b)/(b - a) = (F_X(b) - F_X(a))/(b - a)$$

Density of X in the proximity of x

$$P(x \leq X \leq x+dx)/dx = (F_X(x+dx) - F_X(x))/dx$$

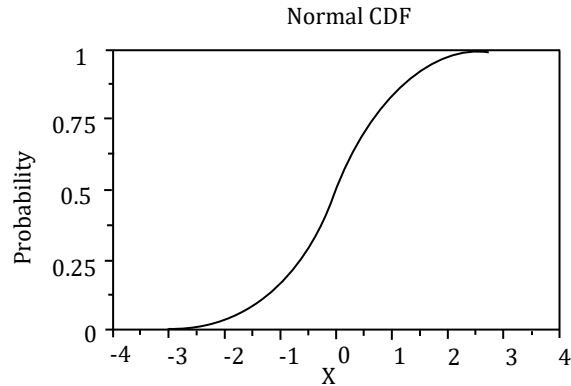
$$f_X(x) = \lim_{dx \rightarrow 0} (F_X(x+dx) - F_X(x))/dx = F_X'(x)$$



Cumulative Distribution Function (Both discrete and continuous)

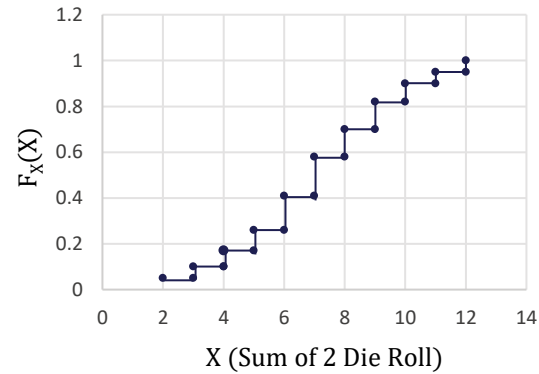
$$F_X(x) = P(X \leq x)$$

E.g.,:- Continuous



E.g.,:- Discrete

Cumulative Distribution Function



$$F_X(3) = 3/36$$

$$F_X(7) = 21/36$$

$$F_X(11.5) = F_X(11) = 35/36$$

Joint RV(Discrete)

$$F(x, y) = P\{X \leq x, Y \leq y\}$$

$$\begin{aligned} F_X(x) &= P\{X \leq x\} \\ &= P\{X \leq x, Y < \infty\} \\ &= F\{x, \infty\} \end{aligned}$$

$$P\{X = x_i\} = (U_j\{X = x_i, Y = y_j\})$$

$$= \sum_j P\{X = x_i, Y = y_j\}$$

$$= \sum_j P\{x_i, y_j\}$$

Suppose that 3 batteries are randomly chosen from a group of 3 new, 4 used but still working, and 5 defective batteries



$$p(0,0) = \binom{5}{3} / \binom{12}{3} = 10/220$$

$$p(0,1) = \binom{4}{1} \binom{5}{2} / \binom{12}{3} = 40/220$$

$$p(0,2) = \binom{4}{2} \binom{5}{1} / \binom{12}{3} = 30/220$$

$$p(0,3) = \binom{4}{3} / \binom{12}{3} = 4/220$$

$$p(1,0) = \binom{3}{1} \binom{5}{2} / \binom{12}{3} = 30/220$$

$$p(1,1) = \binom{3}{1} \binom{4}{1} \binom{5}{1} / \binom{12}{3} = 60/220$$

i \ j	0	1	2	3	Row Sum =P{X = i}
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column Sums= P{Y = j}	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

X = No. of new batteries selected.

Y = No. of used but working batteries selected.

Joint RV(Continuous)

$$P\{(X, Y) \in C\} = \iint_{(x,y) \in C} f(x,y) dx dy$$

$$P\{X \in A, Y \in B\} = \int_B \int_A f(x,y) dx dy$$

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

$$P\{X \in A\} = P\{X \in A, Y \in (-\infty, \infty)\}$$

$$= \int_A \int_{-\infty}^{\infty} f(x,y) dy dx$$

$$= \int_A f_X(x) dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy$$

Joint RV(Continuous)

The joint density function of X and Y is given by

$$f(x, y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x < \infty, 0 < y < \infty \\ 0 & \text{Otherwise} \end{cases}$$

Compute (a) $P\{X > 1, Y < 1\}$; (b) $P\{X < Y\}$ and (c) $P\{X < a\}$

Jointly Distributed Random Variables

$$\begin{aligned} \text{(a) } P\{X > 1, Y < 1\} &= \int_0^1 \int_1^\infty 2e^{-2y}e^{-x} dx dy \\ &= \int_0^1 2e^{-2y} (-e^{-x}|_1^\infty) dy \\ &= e^{-1} \int_0^1 2e^{-2y} dy \\ &= e^{-1} (1 - e^{-2}) \end{aligned}$$

$$\begin{aligned} \text{(b) } P\{X < Y\} &= \iint_{(x,y); x < y} 2e^{-x}e^{-2y} dx dy \\ &= \int_0^\infty \int_0^y 2e^{-x}e^{-2y} dx dy \\ &= \int_0^\infty 2e^{-2y}(1 - e^{-y}) dy \\ &= \int_0^\infty 2e^{-2y} dy - \int_0^\infty 2e^{-3y} dy \\ &= 1 - \frac{2}{3} \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{(c) } P\{X < a\} &= \int_0^a \int_0^\infty 2e^{-2y}e^{-x} dy dx \\ &= \int_0^a e^{-x} dx \\ &= 1 - e^{-a} \end{aligned}$$

Independent RV

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$$

$$F(a,b) = F_X(a)F_Y(b) \quad \text{for all } a,b$$

$$F(x,y) = F_X(x)F_Y(y) \quad \text{for all } x,y$$

Conditional Distributions

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

The joint density of X and Y is given by

$$f(x, y) = f(x) = \begin{cases} \frac{12}{5}x(2 - x - y) & 0 < x < 1, 0 < y < 1 \\ 0, & \text{Otherwise} \end{cases}$$

Compute the conditional density of X, given that Y = y, where $0 < y < 1$

For $0 < x < 1, 0 < y < 1$, we have

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} \\ &= \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dx} \\ &= \frac{x(2 - x - y)}{\int_0^1 x(2 - x - y) dx} \\ &= \frac{x(2 - x - y)}{\frac{2}{3} - y/2} \\ &= \frac{6x(2 - x - y)}{4 - 3y} \end{aligned}$$

Expectation

$$E[X] = \sum_i x_i P\{X = x_i\}$$

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

$E[g(X, Y)] = \sum_y \sum_x g(x, y)p(x, y)$ in the discrete case

$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f(x, y)dxdy$ in the continuous case

For example, if $g(X, Y) = X + Y$, then in the continuous case,

$$\begin{aligned} E[X + Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y)f(x, y)dxdy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xf(x, y)dxdy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} yf(x, y)dxdy \\ &= E[X] + E[Y] \end{aligned}$$

If $X = b(\text{const.})$ $E[X] = b$

$Y = aX$; $E[Y] = aE[X]$

$Y = aX + b$; $E[Y] = aE[X] + b$

Variance

If X is a random variable with mean μ , then the variance of X , denoted by $\text{Var}(X)$ denoted by σ^2 , is:

$$\text{Var}(X) = E[(X - \mu)^2]$$

An alternative formula for $\text{Var}(X)$ can be derived as follows:

$$\begin{aligned}\text{Var}(X) &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - E[2\mu X] + E[\mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - \mu^2\end{aligned}$$

That is,

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Square root of Variance is known as Standard Deviation, denoted by σ .

A useful identity concerning variance is that for any 2 constants a and b ,
 $\text{Var}(aX + b) = a^2\text{Var}(X)$

Let $\mu = E[X]$ and recall that $E[aX + b] = a\mu + b$. Thus, by the definition of variance, we have

$$\begin{aligned}\text{Var}(aX + b) &= E[(aX + b - E[aX + b])^2] \\ &= E[(aX + b - a\mu - b)^2] \\ &= E[(aX - a\mu)^2] \\ &= E[a^2 (X - \mu)^2] \\ &= a^2 E[(X - \mu)^2]\end{aligned}$$

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

Covariance

The covariance of two random variables X and Y, written $\text{Cov}(X,Y)$ is defined by

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

When μ_X and μ_Y are the means of X and Y, respectively.

A useful expression for $\text{Cov}(X,Y)$ can be obtained by expanding the right side of the definition:

$$\text{Cov}(X, Y) = E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y]$$

$$= E[XY] - \mu_X E[Y] - \mu_Y E[X] + \mu_X \mu_Y$$

$$= E[XY] - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$\text{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^m Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m \text{Cov}(X_i, Y_j)$$

Corollary

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \text{Cov}(X_i, X_j)$$

If X and Y are independent random variables, then

$$\text{Cov}(X, Y) = 0$$

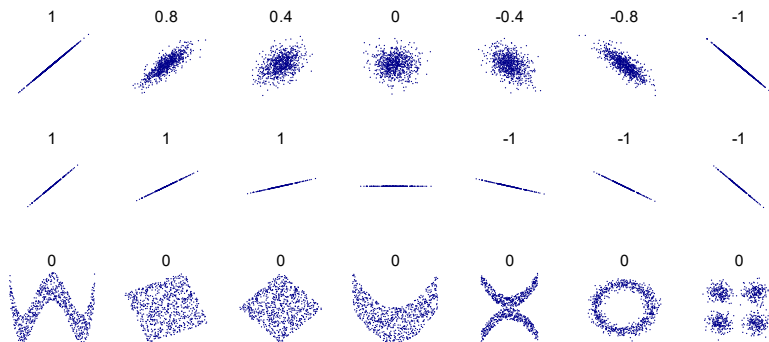
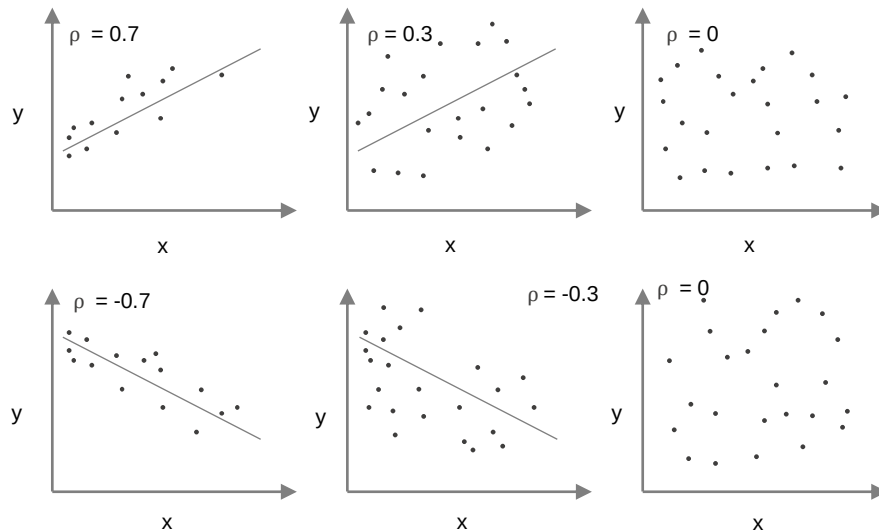
and so, for independent $X_1, \dots, X_n$,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i)$$

Correlation

$$\rho_{X,Y} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

$$\rho_{X,Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - E(X)^2} \cdot \sqrt{E(Y^2) - E(Y)^2}}$$



Covariance Matrix

If there are N Random Variables $X_1, X_2, \dots, X_N$ we study the relationships between them with the help of an $N \times N$ Covariance Matrix.

Let $\mathbf{X} = (X_1, X_2, \dots, X_N)^T$ Then, Covariance Matrix $K_{\mathbf{XX}} = E[(\mathbf{X} - E(\mathbf{X}))^T(\mathbf{X} - E(\mathbf{X}))]$

$$K_{\mathbf{XX}}(i, i) = \text{Var}(X_i)$$

$$K_{\mathbf{XX}}(i, j) = \text{Cov}(X_i, X_j) \text{ for } i \neq j$$

$$K_{\mathbf{XX}} = \begin{bmatrix} E[(X_1 - E[X_1])(X_1 - E[X_1])] & E[(X_1 - E[X_1])(X_2 - E[X_2])] & \cdots & E[(X_1 - E[X_1])(X_n - E[X_n])] \\ E[(X_2 - E[X_2])(X_1 - E[X_1])] & E[(X_2 - E[X_2])(X_2 - E[X_2])] & \cdots & E[(X_2 - E[X_2])(X_n - E[X_n])] \\ \vdots & \vdots & \ddots & \vdots \\ E[(X_n - E[X_n])(X_1 - E[X_1])] & E[(X_n - E[X_n])(X_2 - E[X_2])] & \cdots & E[(X_n - E[X_n])(X_n - E[X_n])] \end{bmatrix}$$

Properties of Covariance Matrix

Covariance Matrix is Symmetric as

$$\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$$

Covariance Matrix has positive trace as

$$\text{Var}(X_i) > 0$$

If X_i 's are mutually independent

$\text{Cov}(X_i, X_j) = 0$ making K_{XX} diagonal matrix.

Special Random Variables – Bernoulli Distribution

Bernoulli trial — a random experiment that has only two outcomes: Success or Failure.

$$P\{X = 1\} = p$$

$$P\{X = 0\} = 1 - p$$

$$E[X] = 1 \cdot P\{X = 1\} + 0 \cdot P\{X = 0\} = p$$

$$\text{Var}[X] = E[X^2] - E[X]^2 =$$

$$(1^2 \cdot p + 0^2 \cdot (1 - p)) - p^2 = p - p^2 = p(1 - p)$$

Eg:- Biased Coin Toss

$$P(H) = 0.6, P(T) = 0.4$$

$$X = 1 \text{ if H else } 0$$

$$E[H] = p = 0.6$$

$$\text{Var}(H) = p - p^2 = 0.24$$



Binomial Distribution

X = No. of Heads when a biased coin is tossed n times

Sample Space of $X = \{0, 1, 2, \dots, n\}$

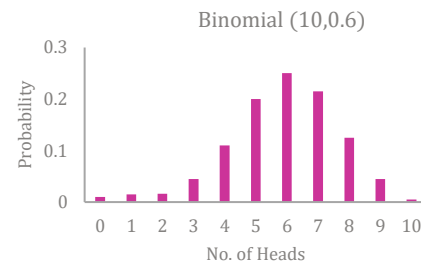
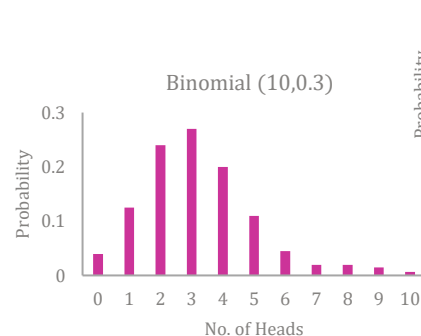
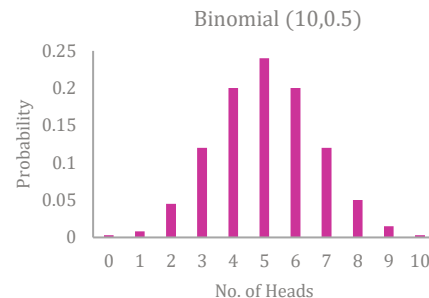
$$P(H) = p; P(T) = 1 - p$$

$$\text{So } P(X=k) = {}^nC_k p^k (1-p)^{n-k}$$

$$X \sim B(n, p)$$

$$E[X] = E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n] = p + \dots + p = np$$

$$\text{Var}(X) = np(1-p)$$



Poisson Distribution

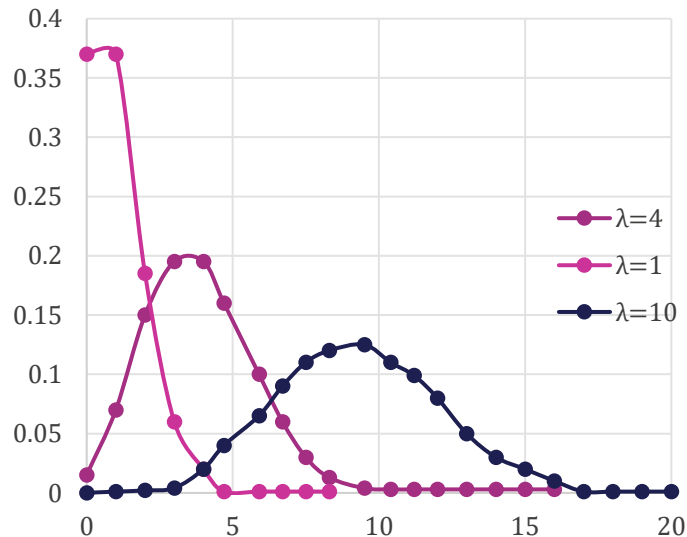
$$X \sim \text{Pois}(\lambda)$$

$$f(k; \lambda) = \Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\lambda = E(X) = \text{Var}(X)$$

$$P(k \text{ events in interval } t) = \frac{(rt)^k e^{-rt}}{k!}$$

Probability mass function



Poisson - Example

A Call Centre receives 180 calls every hour(r).

What is the probability of getting 4 calls in a minute(t) at the call center?

Let X = No. of calls every minute

(i) Events occur randomly

(ii) Mean rate $\lambda = 180 \times \frac{1}{60} = 3$

We can now use the formula to calculate the probability of receiving exactly 4 calls in a minute

$$P(X = 4) = e^{-3} \frac{3^4}{4!} = 0.168$$

Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

μ = Mean

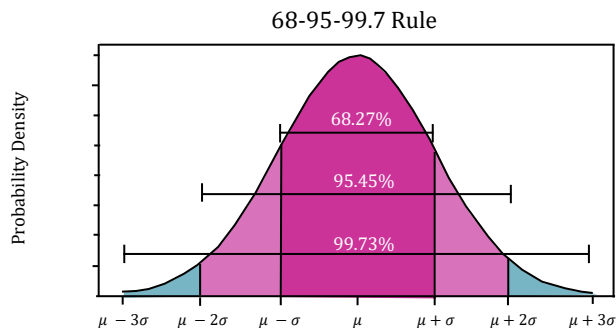
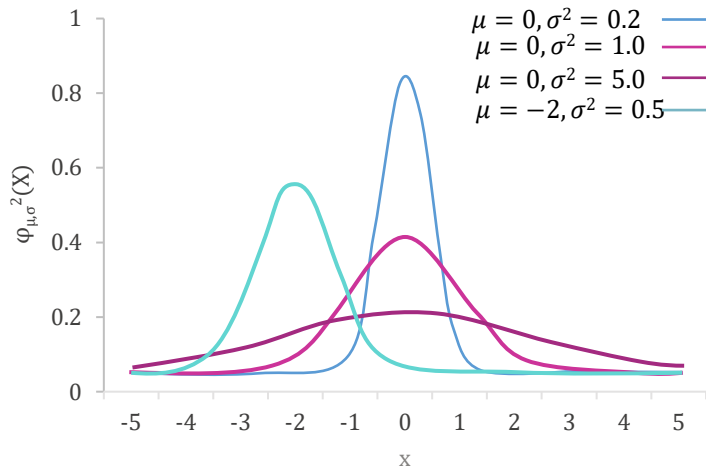
σ^2 = Variance

σ = Standard Deviation

If $X \sim N(\mu, \sigma^2)$, then:

$Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$ where Z is
Standard Normal

Similarly, X could be obtained from
Z, as $X = \sigma Z + \mu$



Multivariate Normal Distribution

$$f(x_1, \dots, x_k) = \frac{\exp(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}))}{\sqrt{(2\pi)^k |\boldsymbol{\Sigma}|}}$$

The multivariate normal distributed of a k-dimensional random vector $\mathbf{x} = (X_1, X_2, \dots, X_k)^T$ can be written in the following notation:

$$\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Or to make it explicitly known that $\mathbf{X}$ is k-dimensional,

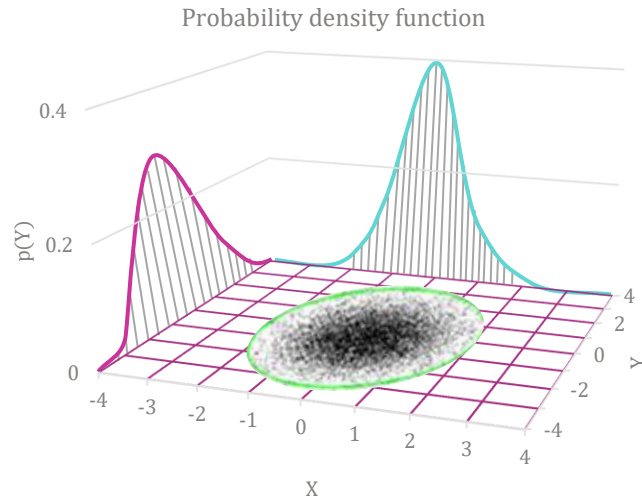
$$\mathbf{X} \sim N_K(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

With K-dimensional **mean vector**

$$\boldsymbol{\mu} = E[\mathbf{X}] = (E[X_1], E[X_2], \dots, E[X_K])^T$$

With K x K **covariance matrix**

$$\boldsymbol{\Sigma} = K_{XX}$$



Many sample points from a multivariate normal distributed with

$$\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{bmatrix} 1 & 3/5 \\ 3/5 & 2 \end{bmatrix}$$

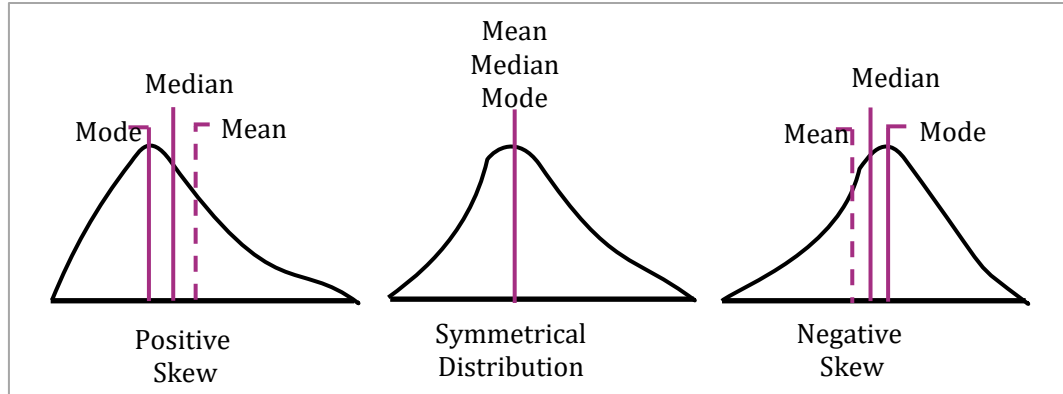
, shown along with the 3-sigma

Ellipse, the two marginal distributions, and the two 1-d histograms

Skewness

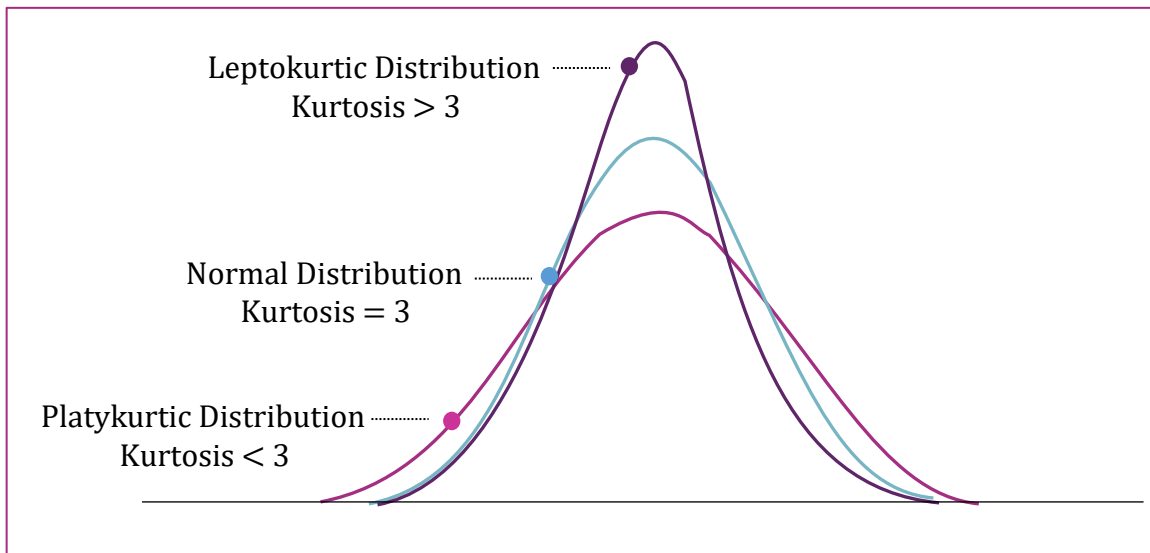
The skewness of a random variable X is the third standardized moment μ_3 , defined as:

$$\bar{\mu}_3 = E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = \frac{\mu_3}{\sigma^3} = \frac{E[(X - \mu)^3]}{(E[(X - \mu)^2])^{\frac{3}{2}}}$$



The Kurtosis is the fourth standardized moment, defined as

$$Kurt[X] = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \frac{E[(X - \mu)^4]}{(E[(X - \mu)^2])^2}$$



Log-Normal Distribution

A positive random variable X is log-normally distributed (i.e., $X \sim \text{Lognormal}(\mu, \sigma^2)$), If the logarithm of X is normally distributed with mean μ and variance σ^2 :

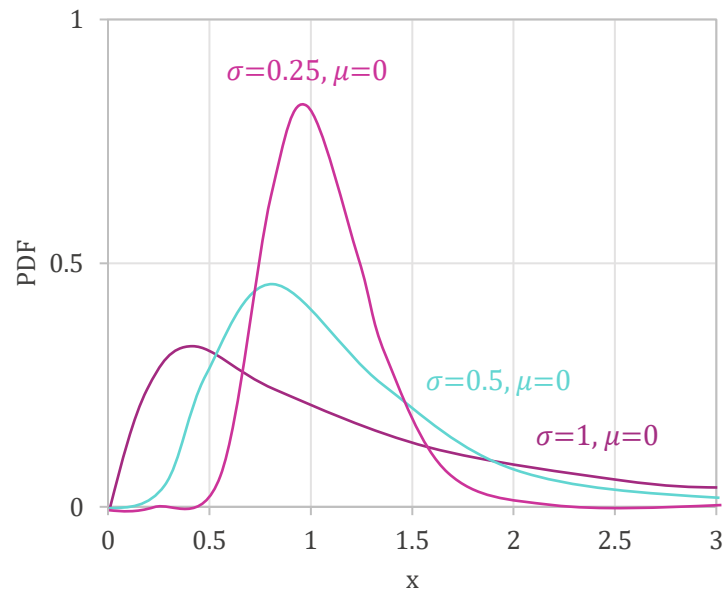
$$\ln(X) \sim N(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$$

$$\text{Mean} \quad \exp\left(\mu + \frac{\sigma^2}{2}\right)$$

$$\text{Variance} \quad [\exp(\sigma^2) - 1] \exp(2\mu + \sigma^2)$$

Probability density function



Miscellaneous Properties

- DeMorgan's Laws for set of events:

$$(A \cup B)^c = A^c \cap B^c$$

More generally:

$$(\cup_n E_i)^c = \cap_n (E_i)^c$$

Similarly:

$$(A \cap B)^c = A^c \cup B^c$$

$$(\cap_n E_i)^c = \cup_n (E_i)^c$$

- If A is independent of B, then:

A^c is independent of B

A is independent of B^c

A^c is independent of B^c

Bayes' Theorem

Bayes Theorem Proof

$$P(E_i|A) = \frac{P(E_i \cap A)}{P(A)}, P(A|E_i) = \frac{P(E_i \cap A)}{P(E_i)}$$

$$P(E_i \cap A) = P(A)P(E_i|A) = P(E_i)P(A|E_i)$$

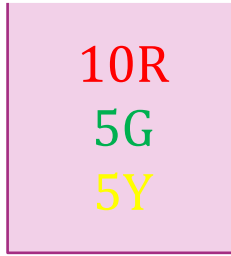
If E_i 's are mutually exclusive and exhaustive

$$P(A) = P(A \cap (\cup_{k=1}^n E_k)) = P(\cup_{k=1}^n (A \cap E_k)) = \sum_{k=1}^n P(A \cap E_k)$$

$$P(A) = \sum_{k=1}^n P(E_k)P(A|E_k)$$

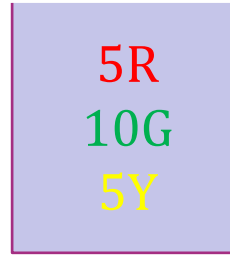
$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{k=1}^n P(E_k)P(A|E_k)}$$

Bayes' Theorem Example



1

$$P(1) = 0.5$$



2

$$P(2) = 0.3$$



3

$$P(3) = 0.2$$

$$\begin{aligned}
 P(1/R) &= \frac{P(R \cap 1)}{P(R)} \\
 &= \frac{P(R/1)P(1)}{P(R/1)P(1) + P(R/2)P(2) + P(R/3)P(3)} \\
 &= \frac{0.5 \times 0.5}{0.5 \times 0.5 + 0.25 \times 0.3 + 0.25 \times 0.2} \\
 &= \frac{2}{3} \approx 0.67
 \end{aligned}$$

Example

Suppose a person is tested for a rare auto-immune disease with a prevalence of 1%. Given that the test has a sensitivity, the probability that the test is positive given the person actually has the disease is 90% and the specificity, or the probability that the test is negative given the person actually doesn't have the disease is 90%; what is the probability that the person has the disease given that the person tested positive?

Given that, $P(+/D) = 0.9$, $P(-/H) = 0.9$, $P(D) = 0.01$, so $P(H) = 0.99$ and $P(+/H) = 0.1$

We now have to compute $P(D/+)$, which is:

$$\begin{aligned} P(D/+) &= \frac{P(+/D)P(D)}{P(+/D)P(D) + P(+/H)P(H)} \\ &= \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.1 \times 0.99} \\ &= \frac{9}{108} = 8.33\% ! \end{aligned}$$

Example

Suppose a person is tested for a rare auto-immune disease with a prevalence of 1%. Given that the test has a sensitivity, the probability that the test is positive given the person actually has the disease is 90% and the specificity, or the probability that the test is negative given the person actually doesn't have the disease is 90%; what is the probability that the person has the disease given that the person tested positive?

Given that, $P(+/D) = 0.9$, $P(-/H) = 0.9$, $P(D) = 0.01$, so $P(H) = 0.99$ and $P(+/H) = 0.1$

Suppose we have a population of 1000, then given a prevalence of 1%, there would actually be 1% of 1000 i.e. 10 people with the disease and 990 would not have the disease. Out of these 10, 90% would be correctly classified as having the disease which is 9. While 1 person will not get identified, making it a False Negative.

We also have 990 people who don't have the disease, but the test would correctly identify only 90% of them i.e. 891 people. So the test would incorrectly identify the 99 people as having the disease even when they don't! These 99 people will be False Positive of the test.

So, the test deems $9 + 99 = 108$ people as having the disease, out of which only 9 actually have the disease and thus the probability of $9/108 = 8.33\%$

Example

Suppose a person is tested for a rare auto-immune disease with a prevalence of 1%. Given that the test has a sensitivity, the probability that the test is positive given the person actually has the disease is 90% and the specificity, or the probability that the test is negative given the person actually doesn't have the disease is 90%; what is the probability that the person has the disease given that the person tested positive twice?

Given that, $P(+/D) = 0.9$, $P(-/H) = 0.9$, $P(+/H) = 0.1$ and $P(-/D) = 0.1$

Since, after the first test $P(D/+)$ was 8.33%, so for this person, the probability of disease would update from 1% to 8.33% thus, $P(D) = 0.0833$ and $P(H) = 0.9166$

We now have to compute $P(D/+)$ for the second test, which is:

$$\begin{aligned} P(D/+) &= \frac{P(+/D)P(D)}{P(+/D)P(D) + P(+/H)P(H)} \\ &= \frac{0.9 \times 0.0833}{0.9 \times 0.0833 + 0.1 \times 0.9166} \end{aligned}$$

~ 45%

What we see is, Posterior = Likelihood * Prior
and, $= \text{Prior}_{t+1} = \text{Posterior}_t$

Appendix: Gamma Distribution

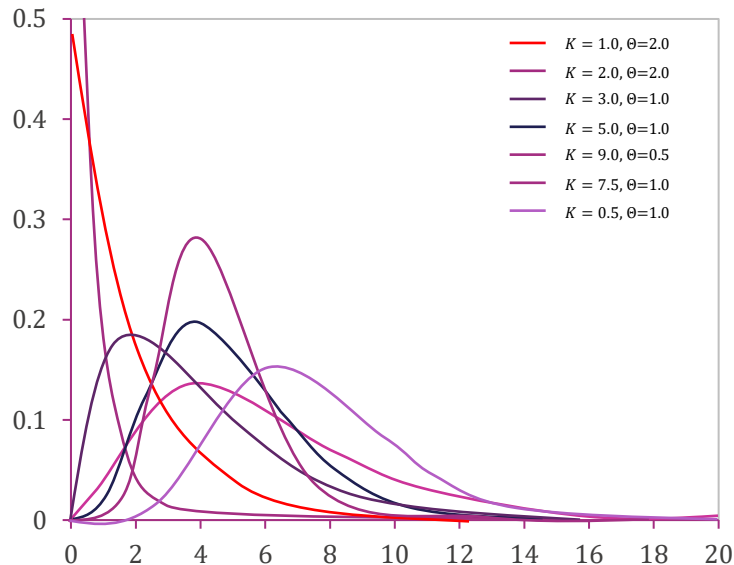
$$X \sim \Gamma(\alpha, \beta) = \text{Gamma}(\alpha, \beta)$$

The corresponding probability density function in the shape-rate parametrization is

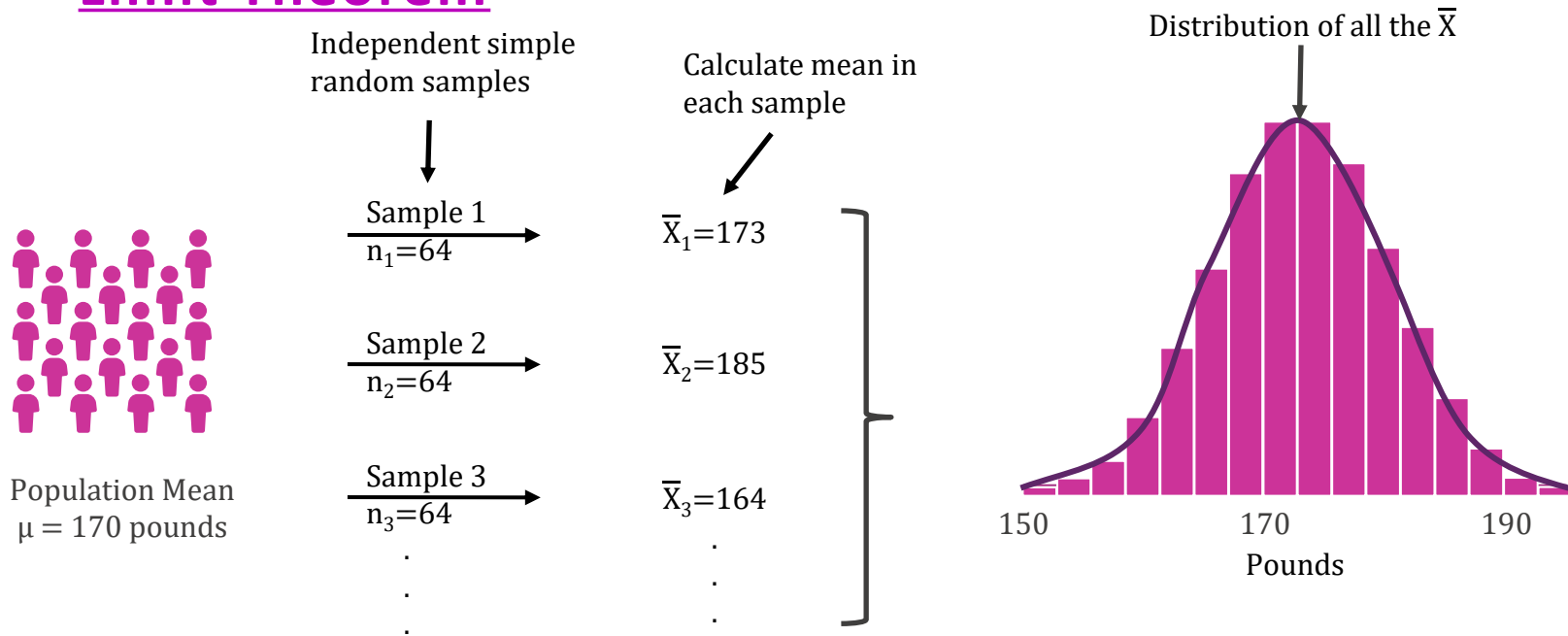
$$f(x; \alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} \text{ for } x > 0 \quad \alpha, \beta > 0,$$

Where $\Gamma(\alpha)$ is the gamma function. For all positive integers, $\Gamma(\alpha) = (\alpha - 1)!$.

Mean	$\frac{\alpha}{\beta}$
Variance	$\frac{\alpha}{\beta^2}$



Sampling Distribution of Mean and Central Limit Theorem

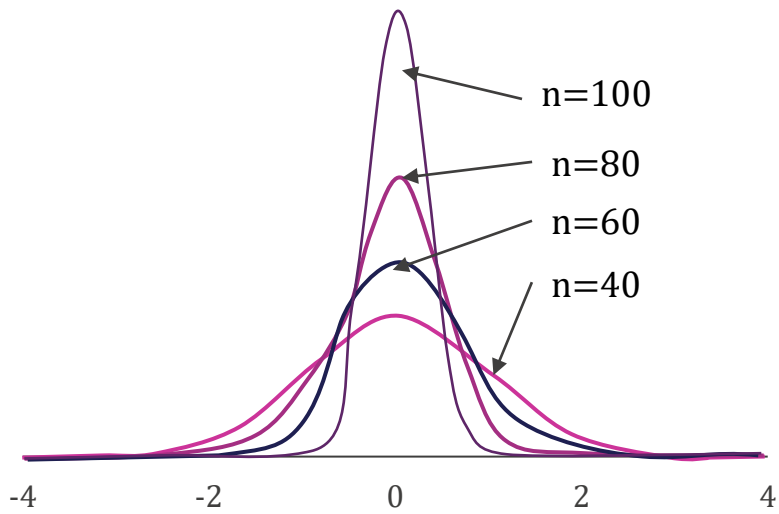


The Central Limit Theorem (CLT)

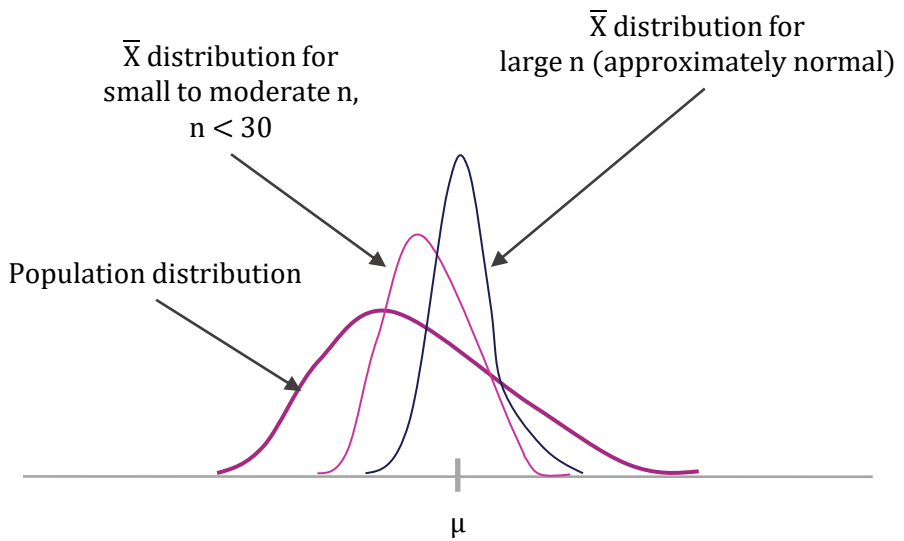
Let $X_1, X_2, \dots, X_m$ be random samples each of size 'n' from a population distribution with mean μ and variance σ^2 . Then if n is sufficiently large (typically $n > 30$):

- The sampling distribution of mean has approximately normal distribution.
- The mean of sampling distribution of mean is same as the population mean i.e. $\mu_{\bar{X}} = \mu$
- The standard deviation of the sampling distribution of mean, or the standard error in the measurement, $\sigma_{\bar{X}} = \sigma/\sqrt{n}$

The larger the value of n, the better the approximation. The distribution of sample means will approximate a Normal distribution regardless of underlying population distribution given that the sample size is greater than 30.



Densities of sample means from a standard normal population.



Densities of population and sample means for small and large n

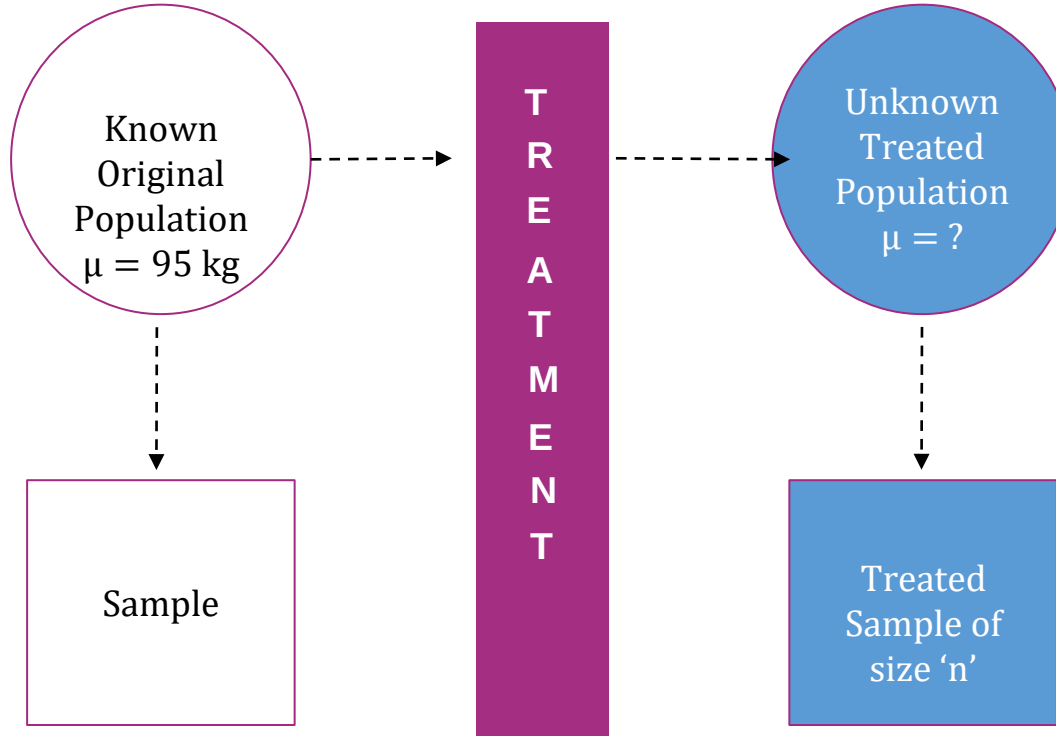
Hypothesis Testing

“Given a sample, if we are to test an apparent effect, what is the probability of seeing such an effect by chance?”

The purpose of the hypothesis test is to decide between two explanations:

1. The difference between the sample and the population can be explained by sampling error (there does not appear to be a treatment effect)
2. The difference between the sample and the population is too large to be explained by sampling error (there does appear to be a treatment effect), thus the effect is statistically significant.

Statistically significant does not always mean that an effect is important or qualitatively significant, it only means that the effect is unlikely to have occurred by chance.



Null and Alternative Hypotheses

Convert the research question to null and alternative hypotheses

- The Null Hypothesis (H_0) claims “apparent effect is not real”.
- The Alternative Hypothesis (H_a) makes claims other than the Null Hypothesis which is to be tested.

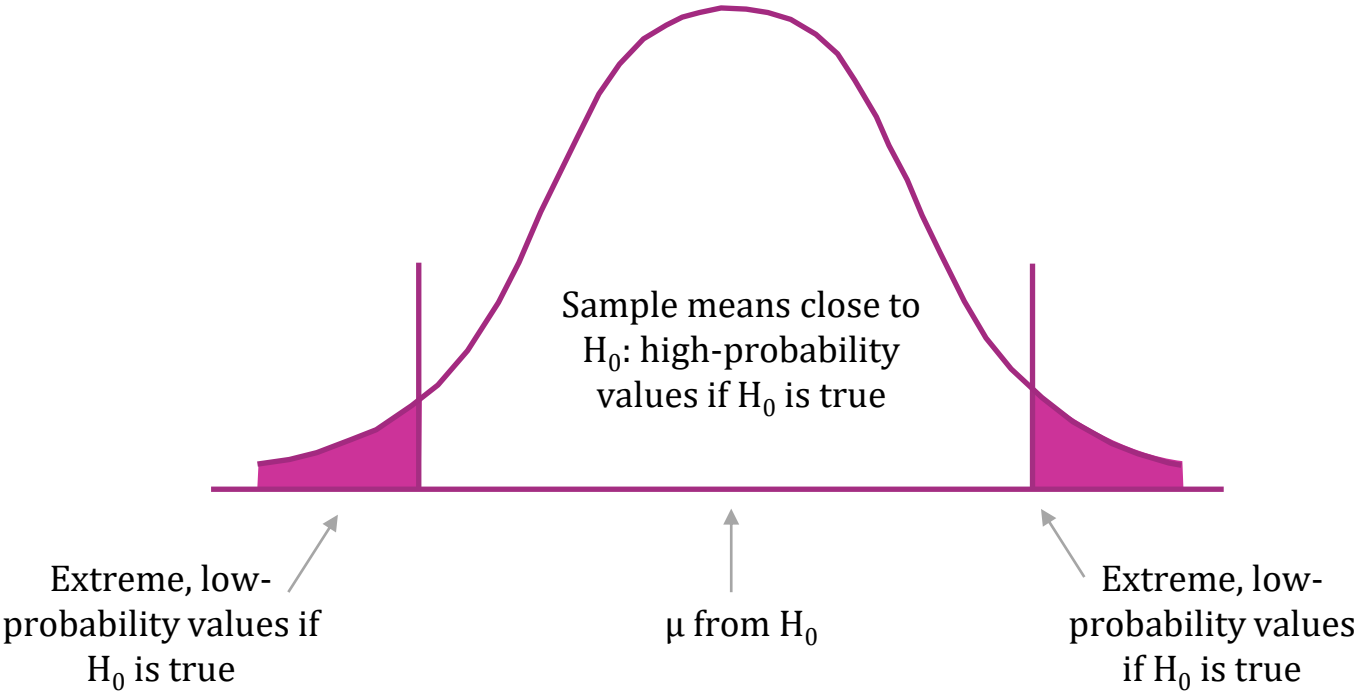
We collect data and seek evidence against H_0 as a way of bolstering H_a

Example:

H_0 : Post exercise, the sample mean is same as population mean, which is 95 kg.

H_a : Post exercise, the sample mean is less than population mean.

The distribution of sample means
if the null hypothesis is true
(all the possible outcomes)



The Matrix of Possibilities

EXPERIMENTER'S DECISION		
	Actual Situation	
	No Effect, H_0 True	Effect Exists, H_0 False
Reject H_0	Type I error	Decision correct
Fail to reject H_0 i.e. Retain H_0	Decision correct	Type II error

Type I Error

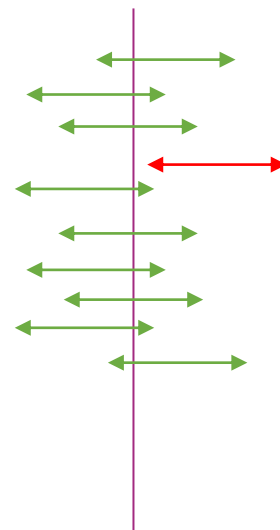
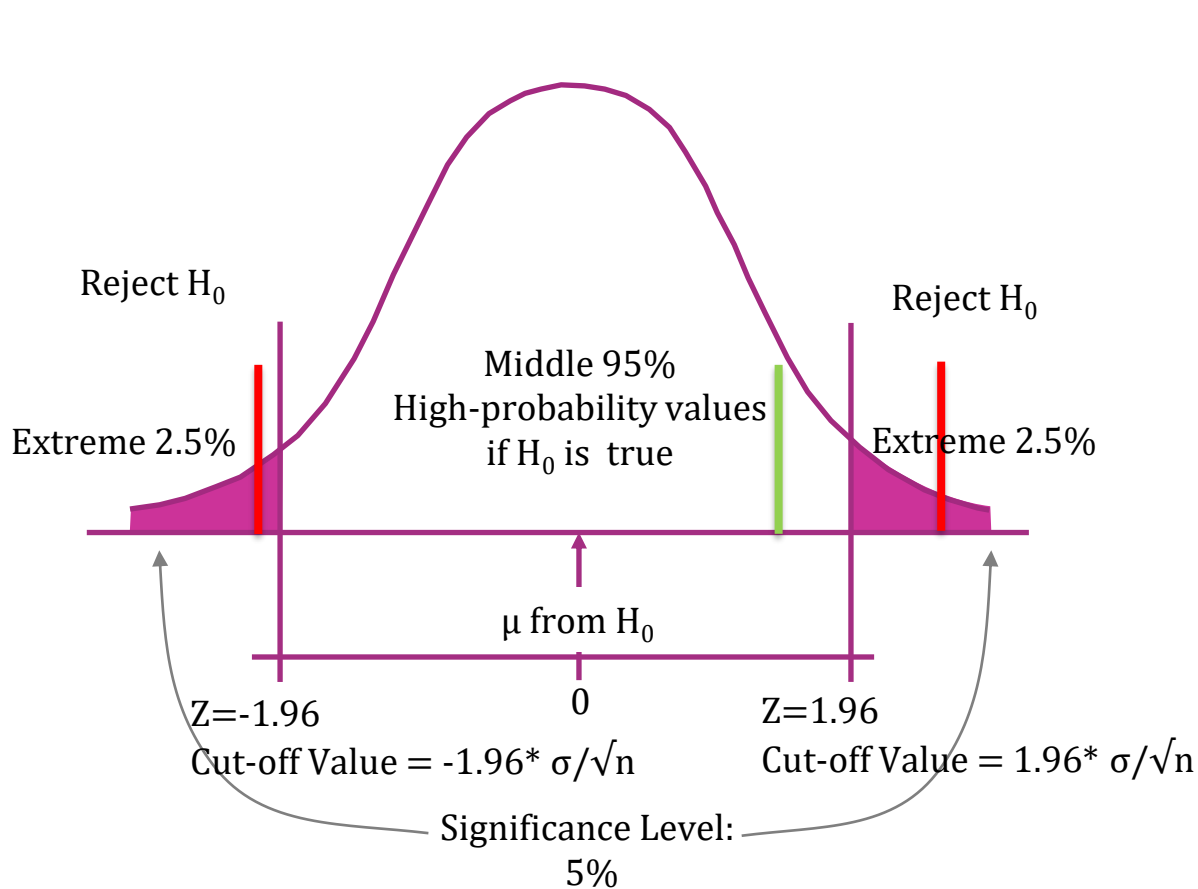
- A Type I error occurs when the sample data appear to show a treatment effect when, in fact, there is none, causing rejection of the null hypothesis and falsely concluding that the treatment has an effect.
- Type I errors are caused by unusual, unrepresentative samples. Just by chance if an extreme sample is selected with the result that the sample falls in the critical region even though the treatment has no effect.
- The hypothesis test is structured so that Type I errors are very unlikely; specifically, the probability of a Type I error is equal to the alpha level.

Type II Error

- A Type II error occurs when the sample does not appear to have been affected by the treatment when, in fact, the treatment does have an effect.
- This causes failure in rejecting the null hypothesis to falsely conclude that the treatment does not have an effect.
- Type II errors are commonly the result of a very small treatment effect. Although the treatment does have an effect, it is not large enough to show up in the research study.

- State the hypotheses and select a Significance level also called as ' α ' level.
- The null hypothesis, H_0 , always states that the treatment has no effect (no change, no difference). According to the null hypothesis, the population mean after treatment is the same as it was before treatment.
- The α level establishes a criterion or cut-off, for making a decision about the null hypothesis. The alpha level also determines the risk of a Type I error.

Confidence Interval – 95%



For a 95% Confidence Interval, the true value will be contained within 95% of such intervals constructed under repeated sampling.

- Locate the critical region. The critical region consists of outcomes that are very unlikely to occur if the null hypothesis is true.
- The critical region is defined by sample means that are nearly impossible to obtain if the treatment has no effect.
- This means that these samples have a probability that is less than the alpha level.

- Compute the test statistic.
- The test statistic forms a ratio comparing the obtained difference between the sample mean and the hypothesized population mean versus the amount of difference we would expect without any treatment effect.
- Compute the p-value, which is the probability of seeing the apparent effect if the null hypothesis is true, which means the probability of seeing a difference as big under the null hypothesis.

Z- Test for Testing Means

Test Condition

Population normal
and infinite

Sample size large
or small

Population
variance is known

H_a may be one-
sided or two sided

Test Statistics

$$Z = \frac{\bar{X} - \mu_{H_0}}{\sigma_p / \sqrt{n}}$$

t- test for testing means

Test Condition

Population is infinite
and normal

Sample size
is small

Population variance
is unknown

H_a may be one-
sided or two sided

Test Statistics

$$t = \frac{\bar{X} - \mu_{H_0}}{\sigma_s / \sqrt{n}}$$

with d. f. = $n - 1$

$$\sigma_s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{(n - 1)}}$$

Z test for testing the difference between the means

Test Condition

Sample happen to be small

Presumed to have been drawn from the same population

Population variance are unknown but assumed to be equal

H_a may be one-sided or two sided

Test Statistics

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{(n_1 - 1)\sigma_{s_1}^2 + (n_2 - 1)\sigma_{s_2}^2}{n_1 + n_2 - 2}} \times \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with d. f. = (n₁ + n₂ - 2)

Paired t-test for comparing 2 related samples

Test Condition

Sample happen to be small

Variances of the two populations need not be equal

Population are normal

H_a may be one-sided or two sided

Test Statistics

$$t = \frac{\bar{D} - 0}{\sigma_{\text{diff.}}/\sqrt{n}}$$

with (n - 1)d. f.

$\bar{D}$ = Mean of differences

$\sigma_{\text{diff.}}$ = Standard deviation of differences

n = Number of matched pairs

- Large value for the test statistic shows that the obtained mean difference is more than would be expected if there is no treatment effect.
- If large enough to be in the critical region, we conclude that the difference is significant or that the treatment has a significant effect, in which case we reject the null hypothesis.
- If the mean difference is relatively small, then the test statistic will have a low value. In this case, we conclude that the evidence from the sample is not sufficient, and the decision is failure to reject the null hypothesis.

Measuring Effect Size

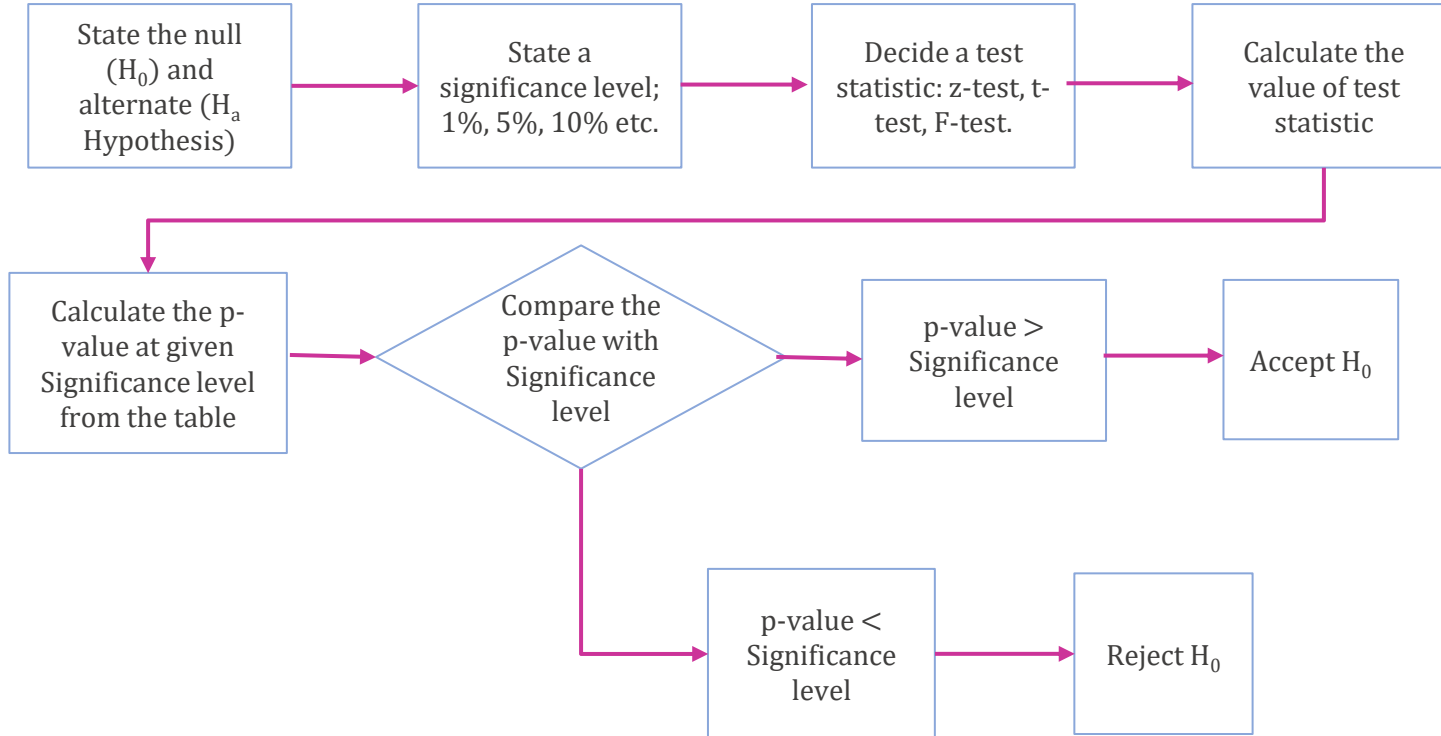
A hypothesis test evaluates the statistical significance of the results from a study. The test determines whether or not it is likely that the obtained sample mean occurred without any contribution from a treatment effect.

The hypothesis test is influenced not only by the size of the treatment effect but also by the size of the sample.

Thus, even a very small effect can be significant if it is observed in a very large sample.

Because a significant effect does not necessarily mean a large effect, it is recommended that the hypothesis test be accompanied by a measure of the effect size. Cohen's d measures the size of the mean difference in terms of the standard deviation

Hypothesis Testing Flow Diagram



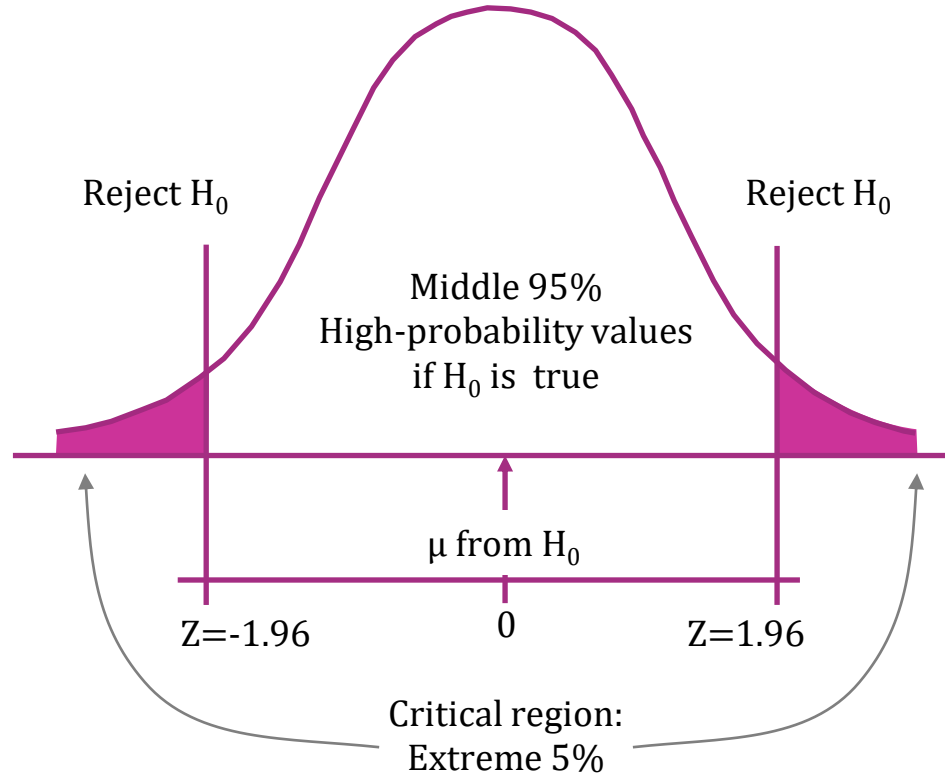
A test is known to have a mean score of 68 and a standard deviation of 13 following approximately normal distribution. A class of 19 students takes the test and has a mean score of 65. Is the class typical of others who have taken the test? Assume a significance level of $p < 0.05$.

Null Hypothesis $H_0: \mu = 68$
Alternative Hypothesis: $H_a: \mu \neq 68$

Computing the z- score:
$$z = \frac{65 - 68}{\frac{13}{\sqrt{19}}} = \frac{-3}{2.982} = -1.006$$

The p-value for a z-score of -1.006 is 0.314416, which is larger than significance level of 0.05. Thus, the Null Hypothesis cannot be rejected.

We can then, conclude that there is not evidence that this class can be considered different from others who have taken the test.



Appendix: Independent RV Example

Suppose that X and Y are independent random variables having the common density function

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{Otherwise} \end{cases}$$

Find the density function of the random variable X/Y .

Jointly Distributed Random Variables

We start by determining the distribution function of X/Y . For $a > 0$

$$F_{\frac{X}{Y}}(a) = P(X/Y \leq a)$$

$$= \iint_{x/y \leq a} e^{-x} e^{-y} dx dy$$

$$= \int_0^{\infty} \int_0^{ay} e^{-x} e^{-y} dx dy$$

$$= \int_0^{\infty} (1 - e^{-ay}) e^{-y} dy$$

$$= \left[-e^{-y} + \frac{e^{-(a+1)y}}{a+1} \right] \Big|_0^{\infty}$$

$$= 1 - \frac{1}{a+1}$$

$$f_{\frac{X}{Y}}(a) = F'_{\frac{X}{Y}}(a) = \frac{1}{(a+1)^2}$$

Appendix: Normal Distribution

Example

Suppose that a binary message – either “0” or “1” – must be transmitted by wire from location A to location B. However, the data sent over the wire are subject to a channel noise disturbance and so to reduce the possibility of error, the value 2 is sent over the wire when the message is “1” and the value – 2 is sent when the message is “0”. If $x, x = \pm 2$ is the value sent at location A the R , the value received at location B, is given by $R = x + N$, where N is the channel noise disturbance. When the message is received at location B, the receiver decodes it according to the following rule:

if $R \geq .5$, then 1 is concluded
if $R < .5$, then “0” is concluded

Because the channel noise is often normally distributed, we will determine the error probabilities when N is a standard normal random variable.

There are two types of errors that can occur: One is that the message “1” can be incorrectly concluded to be “0” and the other that “0” is incorrectly concluded to be “1.” The first type of error will occur if the message is “1” and $2 + N < .5$, whereas the second will occur if the message is “0” and $-2 + N \geq .5$.

Hence

$$P\{\text{error} | \text{message is "0"}\} = P\{N < -2.5\}$$

And

$$= 1 - \phi(2.5) = .0062$$

